→ return the smallest word in the string.

logic:

- traversing through the sentence.
- keep a variable count and count the number of letters untill a space is encountered
- keep a variable like minimum and compare the length.
- if its small then update it there and store the index to a variable index and then at the end print the value with the help of the index.

```
public static void main(String args[]){

  String smallest=" ";
  String current = " ";

for(int i=0; i<s.length();i++){
    char ch = s.charAt(i);

  if (ch!=' ') {
        current += ch;
  }
  else {
      if (current.length() >0) {
          if (smallest.equals(" ") || current.length
            <smallest.length()) {

              smallest = current;
          }

      current = "";
  }
}
}
if (current.length()>0) {
    if (smallest.equals(" ") || current.length
      < smallest.length
```

```
smallest = current;
}
}
return smallest;
}
}
```

~~return the sent~~

→ Zigzag

- we have numRows rows
- we have to move:
  1) down
  2) then up diagonally
- keep the process running until the string is completed.

```
Public sting conversion( string s, int numRows ){

  if (numRows == 1 || s.length() <= numRows ){
        return s;
  }
  StringBuilder[] rows = new StringBuilder[numRows];

for (int i=0; i< numRows; i++) {
      rows [i] = new StringBuilder();
}
int row = 0;
boolean goingDown = true;

for(char ch : s.toCharArray()){
      rows[row].append(ch);
if(row == 0){
      goingDown = true;
}
else if (row == numRows -1){
      goingDown = false;
}
```

```
        if (goingDown) {
            row++;
        }
        else {
            row--;
        }
    }
    StringBuilder result = new StringBuilder();

    for (StringBuilder sb : rows) {
        result.append(sb);
    }
    return result.toString();
    }
}
```